

Tan Dai Ngo

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EDUCATION

University of Victoria | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for Machine Learning, Systems for Massive Datasets, Data Mining. **GPA:** 8.78 / 9.00.

University of Chicago | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Distributed Systems, Databases, Algorithms, Advanced Python Programming, Web Development.

University of Washington | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Applied Mathematics, Informatics. **GPA:** 3.57 / 4.00. **Certificate:** Quantitative Managerial Economics.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Built REST endpoints on 5+ Java and Spring Boot roaming services, holding Cassandra reads under 500 ms.
- Traced 20+ cross service defects to their root cause and shipped fixes via Jenkins across 50+ releases.
- Debugged production issues with Splunk and built a Mobileum email notifier over the Microsoft Graph API.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.
- Led coding projects focused on debugging, algorithmic reasoning and software performance optimization.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, XGBoost, PyTorch *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six classifiers on 206,000 flows and refit preprocessing on training rows, changing the winner.
- Calibrated binary XGBoost to a 1% false positive budget at 99.5% recall, 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed session leakage as binary PR-AUC fell from 0.981 to 0.125.

Ocean Data ML Platform with RAG | Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10.8 million sensor records into Bronze, then Spark Silver processing into a 3.69M row Gold table.
- Served inference and plain English retrieval behind FastAPI and Qdrant, at hit@k 0.9 and term recall 0.85.
- Benchmarked an air temperature positive control against persistence, cutting RMSE 48% twelve hours out.

Fuel Blending Prediction Service | Python, CatBoost, FastAPI, AWS *Shell.ai Hackathon 2025 | Jul 2025*

- Engineered weighted aggregates, spread and fraction entropy across 10 properties of five component blends.
- Selected multi output CatBoost on out of fold MAPE of 0.64 across five folds, against XGBoost at 1.29.
- Packaged the fitted scaler and training medians alongside the model so serving matched training exactly.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine over 60 minute windows with per market thresholds set in basis points.
- Added a trend strategy needing the 8 period EMA over the 20, price over the 200, and returns clear of fees.
- Backtested both strategies on four months of Coinbase data, then deployed to EC2 with CSV risk controls.

TECHNICAL SKILLS

Languages & Programming: Python, Java, SQL, Go, JavaScript, Cypher, MATLAB, object oriented programming

Machine Learning & Evaluation: PyTorch, XGBoost, CatBoost, scikit-learn, pandas, NumPy, Optuna, MLflow

Data, Retrieval & Messaging: Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

Backend, Cloud, CI/CD & Testing: Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda), Jenkins, pytest

Credentials: Google Advanced Data Analytics (2025); Shell.ai Hackathon 2025; IMC 64 Bids, top 250 of 1,500